

Amanda Stoffers

PHD STUDENT IN ASTROPHYSICS - DATA INTENSIVE SCIENCE

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Research Interests

My work centres on scalable probabilistic modelling and GPU-accelerated Bayesian inference, with a particular interest in gradient-based sampling methods, hyperparameter optimisation strategies, and the development of differentiable data analysis pipelines. I am currently building a JAX-based framework for spectral energy distribution (SED) fitting as part of the Data Intensive Science CDT, enabling efficient inference on large astrophysical datasets. Scientifically, my research focuses on the formation and evolution of galaxies in the early Universe, especially the Epoch of Reionization and the mechanisms regulating the escape of ionizing radiation. This includes stellar population synthesis, joint spectroscopic-photometric modelling, and hierarchical approaches to constraining physical galaxy properties.

Education

- 2023 – Present **PhD in Physics** *University of Cambridge* - Cambridge, UK
Member of the Galaxy Formation and Evolution Group at the Kavli Institute for Cosmology.
Supervisor: Dr. Sandro Tacchella.
- 2023 – 2024 **Graduate Training Programme in Data Intensive Science** *University of Cambridge* - Cambridge, UK
Intensive coursework in statistics, machine learning, and scientific computing forming the foundation for SED modelling research.
- 2022 – 2023 **Master of Advanced Studies (Physics)** *University of Cambridge* - Cambridge, UK
Research project on quantifying the escape fraction of ionizing photons received First Class Honours. Graduated top 10 in cohort.
- 2018 – 2022 **B.Sc. in Physics (Grade: 1.3)** *Technische Universität Berlin* - Berlin, Germany
Bachelor thesis: *Quantum theory of electron-phonon interaction in gold nanoparticles* (Grade: 1.0).
- 2018 – 2019 **Promotion of Young Ensemble Leaders** *Landesmusikrat Berlin* - Berlin, Germany
Training in rehearsal management, conducting, and music theory.
- 2016 – 2018 **Abitur (Grade: 1.0)** *Hildegard-Wegscheider Gymnasium* - Berlin, Germany
Included participation in Advanced Physics AG at TU Berlin.

Scholarships

- 2023 – 2027 **Data Intensive Science / Cavendish Scholarship** *STFC* - Cambridge, UK
- 2021 – 2023 **Scholarship of the German Academic Scholarship Foundation (Studienstiftung)**
- 2021 **Enable AI Scholarship** *Data Science and Machine Learning Training*
- 2017 **KMK Scholarship** *Exchange Programme, Trentino, Italy*
- 2014 – 2023 **Scholarship of the Music School City West** *Piano studies*

Awards and Grants

- 2025 **ESA-funded research visit** *JADES Science Focus Meeting* - Santa Cruz, California, USA

- 2025 **Postgraduate Research Fund** *Jesus College, University of Cambridge* - Port Douglas, Australia
Funding to present research at the “Kaba Kada: Exploring the First Billion Years of the Universe” conference.
- 2025 **Winner – Best Talk and Presentation** *Graduate Research Conference* - Jesus College, Cambridge
- 2025 **Research Culture Grant** *Kavli Institute for Cosmology* - Cambridge, UK
Support for the organisation of the Astro Career Talk Series.
- 2018 **Mathematics Abitur Award** *Deutsche Mathematiker Vereinigung* - Berlin, Germany
- 2014 **1st Place, Mathematics Olympiad (Regional Round)** - Berlin, Germany

Work Experience

- 2023 **Quantum Software Engineer** *Diehl Defence* - Überlingen, Germany
Implementation of quantum algorithms and development of mathematical models.
- 2021 – 2022 **Student Tutor** *DESY Zeuthen (German Electron Synchrotron)* - Zeuthen, Germany
Supervision of experiments in the student laboratory.
- 2021 – 2022 **Educational Assistant** *Grunewald Primary School* - Berlin, Germany
Support in classroom settings and learning supervision.
- 2018 – Present **Moderator and Event Assistant** *Planetarium Berlin* - Berlin, Germany
Giving astronomy talks, event supervision, public science communication.

Publications

First-author Publications

- 2025 **The Challenge in Illuminating the Invisible: Constraining LyC Escape with Bayesian Modelling and Symbolic Regression** - Stoffers, A., Tacchella, S., Simmonds, C., Johnson, B. D., & Maiolino, R., arXiv:2511.02908, accepted to MNRAS in May 2026, ADS link

Co-authored Publications

- 2025 **Bursting at the seams: the star-forming main sequence and its scatter at $z = 3-9$ using NIRCam photometry from JADES** - Simmonds, C., Tacchella, S., McClymont, W., Curtis-Lake, E., D'Eugenio, F., Hainline, K., Johnson, B. D., Kravtsov, A., Puskás, D., Robertson, B., Stoffers, A., Willott, C., Baker, W. M., Belokurov, V. A., Bhatawdekar, R., Bunker, A. J., Carniani, S., Chevallard, J., Curti, M., Duan, Q., Helton, J. M., Ji, Z., Looser, T. J., Maiolino, R., Maseda, M. V., Shivaiei, I., Williams, C. C. arXiv:2508.04410. ADS link.
- 2024 **Ionizing properties of galaxies in JADES for a stellar mass complete sample: resolving the cosmic ionizing photon budget crisis at the Epoch of Reionization** - Simmonds, C., Tacchella, S., Hainline, K., Johnson, B. D., Puskás, D., Robertson, B., Baker, W. M., Bhatawdekar, R., Boyett, K., Bunker, A. J., Cargile, P. A., Carniani, S., Chevallard, J., Curti, M., Curtis-Lake, E., Ji, Z., Jones, G. C., Kumari, N., Laseter, I., Maiolino, R., Maseda, M. V., Rinaldi, P., Stoffers, A., Übler, H., Villanueva, N. C., Williams, C. C., Willott, C., Witstok, J., Zhu, Y. *Monthly Notices of the Royal Astronomical Society*, 535(4), 2998–3019. ADS link.

Conference Talks and Presentations

- 2025 **Invited Talk** *JADES Science Focus Meeting* - Santa Cruz, California, USA
CERIDWEN: A GPU-Accelerated Framework for Bayesian SED Fitting.
- 2025 **Conference Talk** *LyC Escape from Galactic Labyrinths* - Crete, Greece
Inferring Lyman Continuum Escape Fractions via SED Modelling.
- 2025 **Conference Talk** *Kaba Kada: Exploring the First Billion Years of the Universe* - Port Douglas, Australia
The Challenge in Illuminating the Invisible: Constraining LyC Escape with Bayesian Modelling and Symbolic Regression.
- 2025 **Workshop Presentation** *IAIFI Summer School and Workshop* - Boston, USA
Contextual Bandits for Stroke Treatment Analysis.
- 2024 **Workshop Presentation** *IAIFI Summer School and Workshop* - Boston, USA
Anomaly Detection via Latent Space Evaluations.
- 2024 **Conference Talk** *National Astronomy Meeting (NAM)* - Hull, UK
Lyman Continuum Escape Fractions of High-Redshift Galaxies.

Community and Professional Activities

- 2026 **Local Organising Committee** *Hubble and Beyond: Morphology and Kinematics of Early Galaxies* Kavli Institute for Cosmology, Cambridge
- 2024 – 2025 **Committee Member** *Cambridge University Women in Physics Society*
- 2024 – 2025 **Committee Member, Astro Career Talk Series** *Kavli Institute for Cosmology, Cambridge*
Organising guest interviews with academic and industry researchers.
- 2024 **Local Organising Committee** *Star Formation 2024* Kavli Institute for Cosmology, Cambridge
- 2022 – 2023 **Student Representative** *MASt Physics Programme, University of Cambridge*
- 2021 – 2025 **Board Member** *In.Ph.A.S.E. Student Association*
Organisation of student colloquia.
- 2019 – 2022 **Volunteer Instructor** *Observatories e.V.*
Tutoring and supervision of experiments in public observatory programmes.

Technical Skills

- Programming Python (JAX, Qiskit, QuTiP, Prospector), C/C++, Linux, L^AT_EX
- Languages German (native), English (near-native), French (B2), Italian (A2), Spanish (A1)
- Other Rowing (W1 Captain, Jesus College, Cambridge); Piano (advanced)